

VAIBHAV KUMAR

Security Automation & QA Engineer | Application Security

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PROFESSIONAL SUMMARY

Security-focused Software & QA Automation Engineer with hands-on experience in security automation, API testing, back-end development, and vulnerability detection. CompTIA Security+ (SY0-701) certified with a strong foundation in threat-detection pipelines, secure API validation, and test engineering across the SDLC/STLC. Skilled in Python, JavaScript/Node.js, Postman, and MITRE ATT&CK-driven detection.

TECHNICAL SKILLS

Security & AppSec: CompTIA Security+ (SY0-701), MITRE ATT&CK, Vulnerability Detection, Threat-Detection Pipelines, Security Automation, Dark Web / OSINT Monitoring, Secure API Validation

Programming: Python, JavaScript (Node.js), C++, Java

Testing & QA: Manual Testing, Test Cases, Test Plans, Bug/Defect Reporting, API Testing, Postman, SDLC, STLC, JIRA

Back-end & APIs: Node.js, Express.js, React.js, REST APIs, Webhooks, HTTP, JSON, Microservices

Databases & Tools: MongoDB, Git, VS Code, OpenCV, LLM APIs

PROFESSIONAL EXPERIENCE

Backend Developer | LyfeIndex Limited

May 2025 – Dec 2025

- Developed and optimized back-end services in Python, improving API response times by 30–40% through efficient database queries and caching strategies.
- Built scalable microservice components handling high-volume transactions, increasing reliability and reducing downtime incidents by 25%.
- Collaborated with front-end and DevOps teams to deliver end-to-end features with seamless API integration and deployment pipelines.

Software Developer Engineer (Security QA) | Phemesoft

Jun 2024 – Dec 2024

- Performed API-level validation and functional testing for internal security workflows using Postman and Python.
- Conducted debugging, defect identification, and issue reporting to improve accuracy of threat-detection pipelines.
- Developed automated dashboards for vulnerability insights; partnered with security, back-end, and data teams to validate system behavior.

KEY PROJECTS

AI-Powered Vulnerability Detection & Testing Engine — Python, Node.js, MongoDB, MITRE ATT&CK

- Built and validated a detection engine identifying suspicious processes; automated threat tagging with MITRE ATT&CK, improving detection accuracy by 35%.
- Wrote unit and functional tests for detection rules and verified performance through simulated attack scenarios.

Dark Web Exposure Monitor — Python, OpenCV, LLM APIs

- Developed automated OSINT scripts to detect leaked data across forums; added data validation and error handling on the scraping pipeline.

API Automation & Testing Suite — Java, Postman, REST APIs, JSON, JIRA

- Designed API test cases for auth, data validation, and CRUD operations; validated response codes, JSON schema, and error conditions.

CERTIFICATIONS

- CompTIA Security+ (SY0-701) • EC-Council Ethical Hacking Essentials (EHE) & Network Defense Essentials (NDE)

EDUCATION

B.Tech, Computer Science — UPES, Dehradun

2021 – 2025 | CGPA: 7.79

LEADERSHIP & ACTIVITIES

- Developer at Amazon HackOn & IIT Jammu Anveshanam '24 hackathons; mentored juniors in security tooling; open-source security/automation contributor.